PRACTICAL-06

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ROLLNO-18

SECTION-B4-B2

AIM--COST ESTIMATION FOR ONLINE FOOD DELIVERY SYSTEM

Python Code -

main.py Visualize Run

```
# COCOMO Cost Estimation
# Project: Online Food Delivery System

print("============================================")
print("     COCOMO COST ESTIMATION")
print("     Online Food Delivery System")
print("============================================")

# Input project size
loc = float(input("Enter estimated Lines of Code (LOC): "))

# Convert LOC into KLOC
kloc = loc / 1000

# Cost per person-month
cost_per_month = float(
    input("Enter cost per person-month in Rs. (e.g. 50000): ")
)

print("\nProject Size:")
print("LOC  :", loc)
print("KLOC :", kloc)

# COCOMO coefficients
```

```
cost_per_month = float(
    input("Enter cost per person-month in Rs. (e.g. 50000): ")
)

print("\nProject Size:")
print("LOC  :", loc)
print("KLOC :", kloc)

# COCOMO coefficients
models = {
    "Organic": (2.4, 1.05, 2.5, 0.38),
    "Semi-Detached": (3.0, 1.12, 2.5, 0.35),
    "Embedded": (3.6, 1.20, 2.5, 0.32)
}

print("\n============================================")
print("           COCOMO ESTIMATION")
print("============================================")

for model, values in models.items():

    a, b, c, d = values
```

```
36
37      a, b, c, d = values
38
39      # Effort in person-months
40      effort = a * (kloc ** b)
41
42      # Development time in months
43      development_time = c * (effort ** d)
44
45      # Average number of people
46      staff = effort / development_time
47
48      # Total estimated cost
49      total_cost = effort * cost_per_month
50
51      print("\nProject Type :", model)
52      print("Effort       :", round(effort, 2), "Person-Months")
53      print("Development  :", round(development_time, 2), "Months")
54      print("Average Staff:", round(staff, 2), "Persons")
55      print("Estimated Cost: Rs.", round(total_cost, 2))
56
57  print("\n================================================")
58  print("              END OF PROGRAM")
59  print("================================================
```

OUTPUT-

```
================================================
     COCOMO COST ESTIMATION
     Online Food Delivery System
================================================
Enter estimated Lines of Code (LOC): 5000
Enter cost per person-month in Rs. (e.g. 50000): 50000

Project Size:
LOC  : 5000.0
KLOC : 5.0

================================================
           COCOMO ESTIMATION
================================================

Project Type : Organic
Effort       : 13.01 Person-Months
Development  : 6.63 Months
Average Staff: 1.96 Persons
Estimated Cost: Rs. 650279.03

Project Type : Semi-Detached
Effort       : 18.2 Person-Months
```

```
============================================
           COCOMO ESTIMATION
============================================

Project Type : Organic
Effort       : 13.01 Person-Months
Development  : 6.63 Months
Average Staff: 1.96 Persons
Estimated Cost: Rs. 650279.03

Project Type : Semi-Detached
Effort       : 18.2 Person-Months
Development  : 6.9 Months
Average Staff: 2.64 Persons
Estimated Cost: Rs. 909782.68

Project Type : Embedded
Effort       : 24.84 Person-Months
Development  : 6.99 Months
Average Staff: 3.55 Persons
Estimated Cost: Rs. 1241756.7
```